

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester VI)
Artificial Intelligence Practical

Practical – IV

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Aim: Bayes' Rule Application

1. Scenario:

A medical test is used to detect a particular disease.

The probability that a randomly selected person has the disease is 1%.

If a person has the disease, the test returns positive with probability 99%.

If a person does not have the disease, the test still returns positive with probability 5% (false positive rate).

A person takes the test and receives a positive result.

Problem Statement:

Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

Code:

#Probability of having disease

P_D = 0.01

#Probability of positive test if disease exists

P_Pos_D = 0.99

#Probability of positive test if disease does not exist

P_Pos_NotD = 0.05

Probability of not having disease

P_NotD = 1-P_D

Bayes Rule

P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))

print("Probability that person actually has disease:")

print(round(P_D_Pos * 100, 2), "%")

print("Divyanjali yadav T052")

Output:

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Probability that person actually has disease:
16.67 %
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